

Relevant Datasets for Digital Phenotyping

For (smartphone and wearable sensing for behavior and mental health)

Executive summary

This report recommends three public datasets that best support a digital phenotyping project, especially for modeling mental health-related outcomes from passive sensing. The selected datasets cover (1) smartphone sensing plus self-reported mental health in a student population (StudentLife), (2) clinically relevant motor-activity sensing for depression status (Depresjon), and (3) large-scale in-the-wild behavioral context recognition from mobile and wearable sensors (ExtraSensory). Together, they enable robust feature engineering, model training, and cross-domain validation.

Selection criteria

- Digital phenotyping fit: passive sensing of behavior (mobility, activity, phone use, social exposure) and/or wearable activity signals.
- Mental health relevance: survey or clinical labels (e.g., stress, depression status) or strong behavioral labels usable as proxies.
- Research usability: clear documentation, stable access, and sufficient sample size or duration.
- Modeling readiness: structured files, repeatable splits (subject-level), and feature/label availability

Recommended datasets

Dataset	Sensors / modalities	Labels / outcomes	Best for
StudentLife (Dartmouth)	Phone sensing (activity, location, phone use, conversation/social) + EMA	Stress/affect EMA; survey outcomes (e.g., PHQ-9, depends on release)	End-to-end phenotyping: features -> mental health prediction
Depresjon (Simula)	Wearable actigraphy (motor activity time series)	Depression vs control (23 patients, 32 controls)	Clinical validation; activity-rhythm depression classification
ExtraSensory (UCSD)	Phone + smartwatch; released as features/labels + optional raw data	Context labels (50+), multi-label per minute; 60 users, 300k+ minutes	Context modeling + feature templates for downstream phenotyping

1) StudentLife (Dartmouth College)

StudentLife is a classic digital phenotyping dataset collected from undergraduate students over an academic term. It combines continuous smartphone sensing with ecological momentary assessments (EMA) and mental-health surveys.

Why it is central: it closely matches the typical digital phenotyping pipeline (passive sensing -> engineered features -> prediction of stress/mood or depression-related measures).

Strengths

- Smartphone sensing plus EMA and survey outcomes (supports supervised learning).
- Term-long longitudinal structure is ideal for temporal features (routines, circadian patterns, weekly cycles).
- Well-studied benchmark: many papers and code examples exist, which helps reproducibility.

Limitations / cautions

- Population is specific (college students), so generalization to other groups should be tested.
- Data may include missingness common to real-world sensing; robust handling is required.
- Some distributions can be term-specific (midterms/finals) and must be accounted for in evaluation splits.

High-value modeling tasks

- Stress level prediction from mobility + phone-use + sleep/activity proxies (daily or weekly).
- Depression severity screening using survey scores as labels (if available in the provided version you use).
- Early-warning change detection (identify rising stress trend before deadlines).

Feature ideas to extract

- Mobility: daily distance, radius of gyration, time-at-home, location entropy.
- Activity/sleep proxies: sedentary vs active duration; inferred sleep duration regularity.
- Phone-use: unlock frequency, screen time, app category usage (if available).
- Social exposure: conversation duration/frequency proxies (non-content).
- Temporal: weekday vs weekend differences; circadian regularity indices.

2) Depresjon (Simula)

Depresjon is a wearable actigraphy dataset focused on motor activity patterns for depression. It includes recordings from depressed patients and healthy controls.

Why it is important: it provides a clinically anchored label (case vs control) and a strong signal modality (activity time series) that complements smartphone-only datasets.

Strengths

- Clear clinical grouping (depressed vs control) enables straightforward classification and benchmarking.
- Simple modality (actigraphy) makes it ideal for time-series baselines and interpretability.
- Public access is straightforward via the dataset portal and Zenodo mirror (commonly cited by researchers).

Limitations / cautions

- Compared to phone-based datasets, it lacks rich context (location, phone use, social signals).
- Small sample sizes are typical, so evaluation must be subject-wise with careful cross-validation.
- Actigraphy alone may not capture all depression phenotypes; combine with other datasets for broader coverage.

High-value modeling tasks

- Depression status classification using daily activity rhythms and variability.
- Anomaly or episode detection (identify low-activity periods).
- Feature transfer: test whether activity rhythm features learned here improve StudentLife predictions.

Feature ideas to extract

- Circadian rhythm: day-night activity contrast, interdaily stability, intradaily variability.
- Statistical: mean, variance, skewness, kurtosis, percentile features over windows (e.g., 30 min, 1 h, 24 h).
- Sleep proxies: longest low-activity interval; regularity of sleep onset/offset (approximate).
- Frequency-domain: dominant periodicity around 24 h and harmonics (FFT-based).

3) ExtraSensory (UC San Diego)

ExtraSensory is a large-scale in-the-wild dataset for behavioral context recognition. It provides sensor-derived features (and optional raw signals) and minute-level multi-label context annotations.

Why it matters for digital phenotyping: context recognition (where/what a person is doing) is a core intermediate step that improves downstream mental-health modeling.

Strengths

- Large scale for a sensing dataset (hundreds of thousands of labeled minutes), enabling robust model training.
- Rich set of context labels (multi-label per minute) supports representation learning and hierarchical behavior modeling.
- Convenient release format (features and labels zip) reduces preprocessing burden.

Limitations / cautions

- Primary labels are behavioral context, not clinical mental-health outcomes.
- Self-reported context labels may be noisy; use label smoothing or robust losses when needed.
- Model evaluation must be person-independent to avoid leakage (subject-wise splits).

High-value modeling tasks

- Train context recognition models (multi-label classification) as a foundation.
- Pretrain a representation model, then fine-tune on StudentLife outcomes (transfer learning).
- Build daily behavior summaries (time in transit, social exposure proxy, sedentary time) for phenotyping featur

Best models to learn mental-health anomaly signatures

Mental-health anomalies in digital phenotyping are typically defined as meaningful deviations from a person-specific baseline (sleep timing shifts, mobility drops, social withdrawal, routine disruption). The most effective modeling strategy is to learn a strong representation of normal behavior and then score deviations.

Model families that work well

- **Self-supervised time-series encoders (best overall starting point):** masked-window Transformers and contrastive encoders learn embeddings from unlabeled sensor windows; detect anomalies in embedding space using simple detectors (e.g., Isolation Forest) or one-class objectives.
- **Forecasting models (best for routine deviation):** TCN or LSTM/GRU predictors learn expected next-window behavior; anomaly score is prediction error aggregated by feature groups (sleep/activity/mobility).
- **Sequence autoencoders (classic, strong baseline):** LSTM/TCN autoencoders and VAEs learn to reconstruct typical patterns; high reconstruction error indicates anomalies and supports signature inspection by feature contribution.
- **Change-point detection (best for 'when did it shift?'):** Bayesian or segmentation-based change-point methods on daily summaries identify dates where behavior regimes change (useful for early warning).
- **Personalized / hierarchical baselines (most important principle):** train per-person baselines or hierarchical models (population + personal effects); always use subject-wise splits and per-user thresholds to avoid identity leakage.

Practical Modeling Stack

- Build daily features (sleep regularity, mobility, phone-use, social exposure proxies) + optional sensor-window embeddings.
- Isolation Forest or one-class scoring on daily features; add change-point detection for 'shift' events.
- For labels (stress/PHQ), train a supervised model on top of embeddings/features for validation and interpretation.

References (public sources)

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5. ExtraSensory Dataset site (UC San Diego). Download and documentation. Available: <https://extrasensory.ucsd.edu/>